CRM · PHASE 1 FEASIBILITY

# Recording sales calls, and turning them into leads

Can two telecallers on ordinary Android phones have their inbound calls recorded, transcribed in Tamil-English, and auto-filed as structured CRM leads? This is what the research found — and, just as important, what it could not confirm.

| PREPARED | FOR | METHOD | EVIDENCE |
|---|---|---|---|
| Jul 2026 | RD Interlock Bricks | 2 deep-research runs | 44 sources · adversarial verify |

THE ONE-LINE FINDING

The recording layer is the entire problem. The AI layer is cheap and effectively solved.

On modern Android, no app you install can record a call — that capability is walled off to system apps, and no user consent or company ownership unlocks it. So the project turns on **one decision**: forward the published number into cloud telephony that records, or buy a phone whose built-in recorder writes to a folder your own app can read.

● VERIFIED Survived 3-vote adversarial checking against a primary source ● UNCONFIRMED Sourced but its verification pass didn't complete ● DEAD END Ruled out on the evidence

## 01 The brief

What the client actually wants to happen, end to end, before any of it can be automated.

Leads arrive as phone calls from organic Instagram videos that print the shop's mobile number. Two telecallers answer on their personal Android phones and, over the call, establish a fixed set of facts. The goal is to capture the call, pull those facts out automatically, and drop a ready-to-review lead into the CRM we are building on the existing Next.js / Prisma / Supabase stack.

### The fields every lead needs

- Place · Name · Type of bricks · Number of bricks
- Type of construction · Cost per brick · Total budget
- Free-text notes, a follow-up date & time, and a "wants a quotation?" flag

Phase 2 — WhatsApp quotation and a nurturing sequence — sits downstream of all this and is scoped only lightly here.

## 02 Why on-phone recording is a wall, not a hurdle

This is the counter-intuitive core of the research. The barrier is technical, not legal — so consent, company-owned phones, and "just sideload our own app" do not move it.

● VERIFIED

**Clean call audio needs a system-only permission.** Capturing the call stream requires `CAPTURE_AUDIO_OUTPUT`, which Android grants only to apps signed into the system partition. There is no user-facing "Allow" for it — a telecaller cannot grant it, and neither can the employer.

src · AOSP permission model, BCR permission table

● VERIFIED

**The best third-party recorder confirms it in its own README.** BCR (Basic Call Recorder) states it works only on "rooted devices or devices running custom firmware," lists stock-unrooted support under *non-features*, is "only known to work on Pixel devices," and — decisively — has no INTERNET permission, so it can never upload what it records.

src · github.com/chenxiaolong/BCR

● VERIFIED

**Sideloading your own signed APK removes the Play-Store ban, not the permission wall.** Google's May-2022 policy against accessibility-based recording governs Play distribution only. But that policy was never the blocker — the OS-level permission is, and it is indifferent to how the app was installed.

src · Android developer docs (all-files access)

● UNCONFIRMED

**Shizuku is a dead end for an unattended phone.** Its ADB-privilege trick does not survive a reboot without root, and its own manual names MIUI, ColorOS and Funtouch — exactly the Indian skins in play — as needing device-specific workarounds. A telecaller phone that silently stops recording after a restart is worse than none.

src · shizuku.rikka.app, ShizuCallRecorder README · verification incomplete

THE TRAP THIS AVOIDS

"Let the telecallers consent and install our APK on company phones" feels like it should work. It doesn't — consent answers a legal question the OS never asked. The permission is refused before consent is ever relevant.

## 03 The two paths that actually work

One is best-practice and rents the hard part; one is buildable in-house and owns it. Both are live options — the choice is decided by two cheap tests, not by argument.

PATH A · BEST PRACTICE

### Forward into cloud telephony

Conditionally forward the published mobile number into a cloud "ExoPhone" that records every leg, then webhooks the audio to the CRM.

WHY IT WINS

- Survives Android updates and telecallers changing phones
- Exotel documents forwarding an existing number in as supported ● VERIFIED
- Recording + webhooks come built in

OPEN CAVEATS

- Per-minute forwarding is billed by the telco (Jio/Airtel/Vi), never on the telephony invoice
- Airtel drops the "forwarding" tone only for a *corporate postpaid* number — a personal prepaid SIM keeps it
- Recording on a leg forwarded from a prepaid SIM is **unconfirmed** — get it in writing from the provider

PATH B · BUILD-IT-YOURSELF

### OEM recorder + our uploader app

Buy a handset whose own dialer auto-records to storage; sideload our app that only watches the folder and uploads — it never tries to record.

WHY IT FITS

- "All files access" is user-grantable and not a system permission ● VERIFIED
- Cheapest to run; entirely our own software
- The OEM dialer is a system app, so recording is legitimate

THE LINCHPIN

- If that OEM writes recordings into its app-private `/Android/data`, no app can read them — even with all-files access ● VERIFIED
- Breaks if an OEM update moves the folder

Not viable, and ruled out: MyOperator (recording starts at ₹5,000/mo and the API/webhooks you need are gated to ₹15,000/mo, billed annually); GSM-gateway / SIM-box hardware; and rooting the telecallers' phones as a supported production setup.

## 04 The AI layer is cheap — and not the risk

Once you have an audio file, transcription and extraction are close to free at this volume. Assume ~100 calls a month of ~5 minutes.

### Transcription (Tamil-English "Tanglish", 8 kHz phone audio)

- **Sarvam Saaras v3** is the strongest India-language option: • VERIFIED ₹30/hour (₹45 with speaker separation), and translate costs the same as transcribe.
- Accuracy on this exact audio is the soft spot: vendor figures are ~19% word-error on a general benchmark, with **no published Tamil-telephony or Tanglish number**. A separate benchmark claims ~14% Tamil on real phone calls • UNCONFIRMED .
- Western engines (Deepgram, GPT-4o transcribe) degrade badly on Tamil phone audio • UNCONFIRMED . And don't "enhance" the audio — neural upscaling *raises* error rates.

### Field extraction

- **Claude Haiku 4.5** with a forced JSON schema pulls the ten fields for roughly **₹0.32–0.65 a call** — about ₹32–65/month, halved again with the Batch API. • VERIFIED
- **Claude has no audio input** • VERIFIED — a separate transcription step is mandatory, not optional. Plan the pipeline as two stages.
- Tamil text costs ~4–5× the tokens of English, which is already priced into the figures above.

DESIGN CONSEQUENCE

At ~14–19% word error, extraction will sometimes be wrong. Create every lead as **needs-review** and let the telecaller correct fields, rather than trusting the machine to file clean leads. A human-in-the-loop screen is a feature, not a fallback.

## 05 Legal & consent — the customer, not the telecaller

The employee's consent covers the employee. The inbound customer — who never agreed to anything — is the exposure. This section was hit hardest by the second run's incomplete verification and needs a dedicated legal read before launch.

- India's **DPDP Act 2023** and its Rules (reportedly notified 14 Nov 2025) make the caller a Data Principal with rights over the recording, including **erasure you must action within 90 days**. ● UNCONFIRMED
- A **purpose-specific consent notice** at the start of the call appears to be required, not merely advisable. ● UNCONFIRMED
- Penalties are tiered into the tens and hundreds of crore with no small-business cap stated. ● UNCONFIRMED

> THE DESIGN DECISION THIS FLIPS
>
> That "this call is being recorded" announcement most people rush to disable? **Leave it on.** It is the cheapest possible consent mechanism for the customer. And build a hard-delete path for a caller's recordings and transcripts into the CRM from day one — not bolted on later.

## 06 WhatsApp nurturing — Phase 2, priced later

Deliberately out of depth here; flagged so it isn't mistaken for solved.

The plan — auto-generate a quotation PDF, send it on WhatsApp, then drip a video at send, another at 24 hours, a follow-up call, and a second follow-up — lands on Meta's per-message pricing (in force since July 2025) and its utility / marketing / authentication template rules. A business-solution provider such as AiSensy, Wati, Interakt or Gupshup is the realistic route. This produced **no verified findings** and cannot be costed from the current evidence — it needs its own research pass before commitments.

## 07 What it costs, per month, for ~100 calls

The AI layer is a rounding error either way. The telephony fee is what pushes Path A past the original ₹2,000 target — and the one-time handset is what keeps Path B under it.

| LINE ITEM | PATH A · CLOUD | PATH B · HANDSET |
|---|---|---|
| Telephony plan, 2 agents, recording + API | 1,500–3,500 | – |
| Per-minute call forwarding (telco) | variable* | – |
| Dedicated recording handset (one-time) | – | ~10,000 |
| Transcription — Sarvam, with diarization | 375 | 375 |
| Extraction — Claude Haiku 4.5 | 32–65 | 32–65 |
| Hosting (already on Supabase/Vercel) | ~0 | ~0 |
| **Monthly run-rate** | **~2,000–4,000 +fwd** | **~400** |

All figures ₹/month unless marked. *Forwarding: ~₹0 on Jio unlimited-voice plans; Airtel/Vi divert rates were not confirmed by an authoritative 2026 tariff and remain an open line. Re-verify every price before quoting the client — LLM and STT pricing move fast.

## 08 How much of this to trust

Two research runs. The first was fully adversarially verified; the second hit a session limit and its verification stage never ran — so a large share of Sections 02–05 is sourced-but-unconfirmed and marked accordingly.

| | |
|---|---|
| ● SOLID | The recording wall, BCR's limits, all-files access, the `/Android/data` block, Exotel forwarding, Claude having no audio, and the AI-layer pricing — all verified against primary sources. |
| ● SOFT | Tamil-telephony accuracy numbers, the DPDP specifics, and the Shizuku failure modes are sourced but were not adversarially checked. Treat as strong leads, not settled facts. |
| ● GAP | Not researched: which exact 2026 handset auto-records to *shared* storage; competitor telephony pricing beyond MyOperator; and the entire WhatsApp Phase-2 costing. |

## 09 What to do next

Two cheap tests decide the architecture. Run them in parallel before writing any pipeline code.

1. **Buy one Redmi handset and check where recordings land.** Enable auto-record, make a call, then `adb shell ls` the recordings path. Shared storage → Path B is live. `/Android/data` → that model is out. One afternoon; decides everything.
2. **Get written confirmation from a cloud-telephony provider** that recording works on a leg forwarded in from a personal prepaid SIM. This is the single unverified assumption Path A rests on.
3. **Commission a short DPDP review** for recording inbound customer calls — consent announcement, retention, and the erasure workflow the CRM must support.
4. **Prototype the AI stage now** — it's independent of the recording choice. Feed a sample Tamil call through Sarvam → Claude Haiku and measure real field accuracy on this business's vocabulary.
5. **Then pick a path** from the two test results and build the ingestion pipeline: audio → Supabase Storage → transcribe → extract → needs-review lead.